

Exp 4: Data modelling and form design for a business application in Zoho Creator.

1. Aim

The objective of this experiment is to design a normalized relational data model and construct a comprehensive low-code business application form—the **Vehicle Service & Repair Request Management System**—in Zoho Creator. This includes defining entity schemas, field data types, data validation rules, mandatory field constraints, and creating associated report views for enterprise record management.

2. Entity-Relationship (ER) Diagram & Data Dictionary

Effective low-code application development begins with structured data modelling. The Vehicle Service & Repair Request Management System models relationships between vehicle owners, their registered vehicles, service requests, and servicing categories.

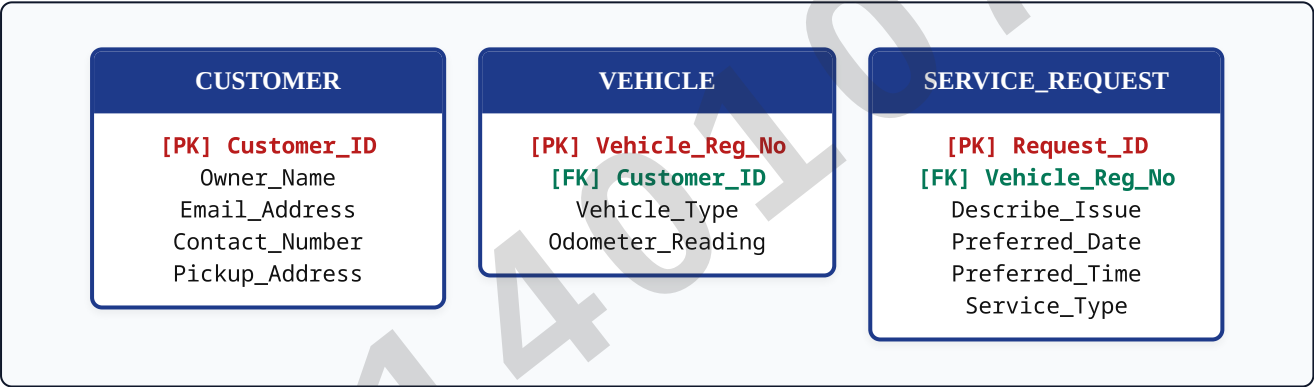


Figure 1: Entity-Relationship (ER) Schema Diagram for Vehicle Service & Repair Management

2.1 Data Dictionary & Schema Specification

Table 1 presents the detailed field definitions, data types, and validation constraints configured in the primary Service_Request_Form in Zoho Creator.

Table 1: Data Dictionary for Service_Request_Form Schema

Field Name	Field Link Name	Data Type	Constraints & Validation	Description
Owner Name	Owner_Name	Name (Composite)	Mandatory (First/Last Name required)	Full name of the vehicle owner submitting the request.
Email Address	Email_Address	Email	Mandatory, Standard RFC 5322 Email Validation	Primary contact email for status updates and invoices.
Pickup Address	Pickup_Address	Address (Composite)	Mandatory (Line 1, City, State, Pincode)	Physical location for vehicle pickup and delivery service.
Contact Number	Contact_Number	Phone	Mandatory, Numeric Format Validation	Mobile phone number for driver/technician coordination.
Vehicle Registration Number	Vehicle_Registration_Number	Single Line	Mandatory, Uppercase Alphanumeric	Unique official registration number of the vehicle.
Describe the Issue	Describe_the_Issue	Multi Line	Optional, Plain Text Box	Detailed breakdown of symptoms, noise, or requested repairs.
Odometer Reading (km)	Odometer_Reading_km	Number	Optional, Positive Integer (≥ 0)	Current distance mileage recorded on vehicle dashboard.
Preferred Service Date	Preferred_Service_Date	Date	Mandatory, Date $\geq$ Current Date	Requested calendar date for maintenance appointment.
Preferred Drop-off Time	Preferred_Drop_off_Time	Time	Optional, Standard Time Picker	Target time window for delivering vehicle to service hub.
Vehicle Type	Vehicle_Type	Drop Down	Options: Hatchback, Sedan, SUV, EV, Commercial	Categorization of vehicle model class for bay allocation.
Service Type	Service_Type	Drop Down	Options: General Service, Periodic Repair, Inspection	Primary scope of maintenance requested by customer.

Field Name	Field Link Name	Data Type	Constraints & Validation	Description
Additional Services Requested	Additional_Services_Requested	Multi Select	Options: Wheel Alignment, Foam Wash, Ceramic Coating	Add-on service options selected by vehicle owner.

3. Procedure

The form design and data model were constructed in Zoho Creator through the following procedure:

- 1. Application Creation:** Logged into the Zoho Creator workspace and initialized a new application titled *"Vehicle Service & Repair Request Management"*.
- 2. Form Canvas Setup:** Navigated to the **DEVELOP > Forms** builder module and instantiated a new form named `Service_Request_Form`.
- 3. Drag-and-Drop Field Assembly:** dragged composite Name, Email, Address, Phone, Number, Single Line, Multi Line, Date, Time, Drop Down, and Multi Select fields from the left-hand tool palette onto the visual canvas.
- 4. Field Property Configuration:** Configured field labels, field link names, default values, placeholder text, and mandatory validation flags in the right-hand Field Properties inspector panel.
- 5. Report View Generation:** Generated an automatic list report titled `Service_Request_Form Report` to enable search, multi-column filtering, and record administrative operations.

4. Output & Screenshots

The following figures document the implemented form schema, runtime form interface, and record report view in Zoho Creator:

Figure 2: Zoho Creator Form Builder Canvas & Field Properties Inspector for Service_Request_Form

Figure 3: Live Application Runtime & User Input Form for Vehicle Service & Repair Requests

The screenshot displays the 'Service_Request_Form Report' view in Zoho Creator. The sidebar on the left contains navigation links for 'Vehicle Service & Repair Request...', 'Service_Request_Form', and 'Service_Request_Form Report'. The main area shows a table of customer records with columns for Owner Name, Email Address, and Pickup Address. A detailed form view is open on the right, showing fields for Owner Name, Email Address, Pickup Address, Contact Number, Vehicle Registration Number, Describe the Issue, Odometer Reading (km), Preferred Service Date, Preferred Drop-off Time, Vehicle Type, Service Type, Additional Services Requested, and Authorization.

Owner Name	Email Address	Pickup Address
Pravinraj P	pravinraj.service@example.com	123 Innovation Way, Tech Park, OMR Road, Karapakkam, Chennai, Tamil Nadu, 600097, India

Owner Name	Pravinraj P
Email Address	pravinraj.service@example.com
Pickup Address	123 Innovation Way, Tech Park, OMR Road, Karapakkam, Chennai, Tamil Nadu, 600097, India
Contact Number	+917010099999
Vehicle Registration Number	TN-01-AB-1234
Describe the Issue	Scheduled 30,000 km periodic service check. Please inspect brake pads and top up engine fluid.
Odometer Reading (km)	30150
Preferred Service Date	18-Aug-2026
Preferred Drop-off Time	10:00:00
Vehicle Type	Hatchback
Service Type	Routine Service
Additional Services Requested	Car Wash Interior Cleaning Tyre Rotation
Authorization	I authorize the service center to prepare an estimate and proceed with the service/repair

Figure 4: Service_Request_Form Report View Displaying Submitted Customer Records

5. Result & Observations

5.1 Observations

- Zoho Creator's visual form builder abstracts database DDL creation, automatically generating normalized table schemas upon form deployment.
- Composite field types (such as Name and Address) simplify data entry by bundling multiple sub-fields into single reusable schema components.
- Built-in validation rules (such as RFC 5322 email syntax and mandatory constraints) ensure high data integrity without writing manual Javascript or SQL validation logic.
- Associated report views are automatically generated alongside form creation, enabling instant record management and multi-criteria searching.

5.2 Result

The data model and user interface form for the **Vehicle Service & Repair Request Management System** were successfully designed, configured, and validated in Zoho Creator in accordance with the experiment objectives.